

GNU Radio: An Open Source Software Radio Ecosystem

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We cannot include software defined radios (SDRs) among the latest concepts evolved in wireless communication. Software radios have been present for more than three decades. However, many processes that were considered theoretical are now practically possible with the rapid pace of growth in digital electronics. Interestingly, a lot of people including hobbyists, academicians and working professionals are increasingly becoming interested in exploiting the opportunities rendered by software radios.

In this article we will focus on GNU Radio, a popular software development toolkit consisting of signal-processing blocks for creating SDRs and other signal-processing systems.

The infrastructure of this software is completely written in C language, whereas many user tools are implemented in Python. The latest stable release of this software licensed under GNU general-public licence (GPL) 3 is version 3.7.5.1, and it is included in this month's DVD for EFY Plus.

Smart graphical interface for easy designs

The first question that might arise in your mind if you are not a very good programmer is, "Do I have to sit and type the entire code?" or "Is there a Simulink-like tool that may make my job easier?"

The answer is yes! This software package comes with GNU Radio Companion (GRC), a graphical signal-processing environment for all simulation needs. The GUI will seem simple and straightforward if you

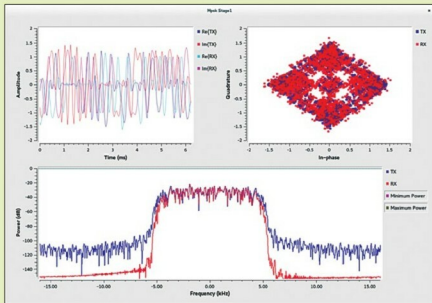


Fig. 1: Plotting demodulated waveform using GNU Radio

Software defined radios

The SDR forum defines these as radios in which some or all physical layer functions are software-defined. The software present in a personal computer or an embedded system can replace hardware components like amplifiers, mixers, modulators/demodulators in a typical radio communication system. Delivery of the final product to the market will be quicker. There will be lesser development costs, minimised maintenance expenses and, ultimately, the end user will get better product at a cheaper rate. Also, since the software is easily replaceable, we can use the same hardware to make different radio systems for various applications.

paid attention in your digital signal-processing classes.

For invoking GRC in Linux, you can call the GRC command. A GRC will pop up on its window and you will have a collection of built-in blocks on the right-hand side of the window to choose from. You can drag these blocks to the main window and connect these by clicking on the edges.

The function of GNU Radio here is to supply you with a multitude of

blocks and execute the flow graphs after their definition. It calls the blocks one after the other, while ensuring that items are passed from one block to another.

How it is different from Simulink

For Simulink users, flow graphs will either run based on frame or on sample. The sample based model is intended to have better control over the signal-processing stream, but in designs that require better